

Using a storyline to increase engagement in a course review

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ABSTRACT

We present an approach to the end of course review session that increases student participation and helps them to identify how the topics that have been covered fit into a broader context. We provide examples of this approach being used for visual computing courses, including the refinements made based on the experience of running them. The approach uses a single strong narrative to connect a series of problems that allow students to apply what they have learnt and, in the process, realise how much they now know and how it fits together. It also serves to remind them what it was that initially attracted them to the topic. We also introduce an element of false jeopardy by pitting groups against each other in a friendly way that further increases the students' engagement in the session.

CCS CONCEPTS

• **Social and professional topics** → *Computing education*; • **Computing methodologies** → *Rendering*; *Computer vision*.

KEYWORDS

Computer vision, rendering, problem solving, narrative, teaching

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1 THE PROBLEM

When teaching a visual computing course, we often find ourselves breaking the content into a series of discrete topics. This is done for the convenience of matching lecture slots and avoiding unnecessary repetition of content between lectures. Or, to simply help students understand ideas in isolation without having to be overly concerned with the bigger picture. Re-introducing the wider context while trying to focus on delivering the detail of a specific topic can be a challenge and is often avoided or only addressed in passing.

It's common practice to use the last lecture of a course to provide a review of or reflection on the content that has been covered. The process has the potential to provide the students with a better understanding of how various ideas taught fit together. However, given that it's not new content, it's tempting for students to skip the

session or not really pay attention. This is especially true where the lecture is purely a summary of the content. In those circumstances, those that do attend are typically only there for any exam hints they might glean.

A less passive approach is to base the review around a formative quiz. We propose taking this approach further by linking the questions via a strong narrative that reintroduces the wider view of the subject. The last session is the ideal time to re-introduce the bigger picture because, the students have covered all the detail of the topics by this stage and we want the students to think about how they might apply their new knowledge.

2 OVERVIEW OF APPROACH

By introducing an overarching narrative, a focus for the review exercise is provided and helps to link the topics covered in the course into a cohesive whole.

We do this by posing a series of open-ended problems that follow the provided narrative and require using one or more of the ideas presented within the course.

Students are divided into groups and given time to discuss and propose solutions to the set of problems. They are expected to consider all the questions.

Groups then take turns presenting their solution to a problem. After one group provides their solution, the floor is opened to the other groups/individuals to suggest any alternatives or make observations about the suggested solution, before the next group moves onto a new question.

3 THE QUESTION ORDER

The following instructions were given to students. They remained on display while the students were discussing the questions.

- The challenge has been broken down into a series of problems.
- All the problems can be addressed using the material that has been covered in the unit.
- Working in small groups look at the problems and decide how you'd tackle them.
- You should consider all the problems.
- Each group in turn will pick a new problem and suggest their solution.
- The order of the groups will be random.

Allowing students to select the question to answer when they are chosen comes with the risk that they will not get their preferred choice of question. It depends on which questions the students before them have answered. Their need to keep re-assessing this perceived risk keeps the students more engaged in the process.

By allowing students to choose the question they answer from the remaining unanswered questions, we move away from the

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narrative order, but gain from an increase in the willingness of the first few groups to provide an answer. This willingness comes from the fact that they are getting to answer the questions for which they believe they have their strongest answer. Which they generally do. Experience also indicates that these are the questions that other groups are most likely to have something else to add to the discussion. Both of these conditions set a trend of participation rather than passive observation that tends to persist through the session, even when the remaining questions have less complete answers.

4 EXAMPLE: COMPUTER VISION

You are being watched. The government has a secret system - a machine that spies on you every hour of every day. I know because...I built it. I designed the machine to detect acts of terror, but it sees everything...

Mr Finch, Person of Interest

The television series, Person of Interest[Nolan 2011], is based on the premise that an AI is able to process CCTV and other data that is maintained on the citizenry. It uses this to predict their behaviour and identify those who are a person of interest. Using a clip of the title sequence from Box of Broadcasts[on Screen 2022] that gives this context, provides a strong story line we use to consider just how the computer vision part of the task can be accomplished. The questions presented are phrased to focus on this and not the artificial intelligence part of the premise.

The story used is of tracking a potential bomber. Each question progresses the story and by providing answers to the questions about how 'the machine' might perform the computer vision tasks, the students are effectively describing how they would build the computer vision components needed and therefore how they can help foil the bomber.

The questions are presented in plot order, but students have tended to choose to answer them out of sequence for the reasons given previously.

We present the last two questions here, to demonstrate how they follow and progress the storyline.

- (7) The machine is concerned that it did not recognise the object in question 6 and it may be a bomb. Fortunately, it has access to a large database of blueprints of devices seen by various government agencies and the military. Assuming it's worked out the geometry of the device it has seen, how might it build models to compare against from the blueprint drawings.
- (8) The machine has decided the unknown object may well be a bomb. It has identified that the person of interest is heading to a popular tourist spot. To assess the level of damage that could be done, a 3D model of the area is needed. Searching the web, the machine finds a large number of photos of the area. How might it use these to construct a 3D model?

The primary aim of the course is 'to understand the ways of analysing images to get information out of them.' The content covered begins with basic image processing, moves on to feature extraction and then using this to perform recognition (both classification and identification) of objects and people, tracking and 3D reconstruction. The course uses face recognition as an example to help students break recognition tasks into identifying a member

Table 1: EETA Metadata

Summary	A narrative links the topics of a course.
Learning Outcomes	Apply content to the wider context.
Classification(s)	Rendering, Modelling, Vision.
Audience	Any.
Dependencies	All course content.
Prerequisites	None.
Strengths	Enjoyable, engaging, helpful.
Weaknesses	None noted.
Variants	Adaptable to other topics.
Assessment	Formative.

of a class or a specific instance. This is reflected in the questions posed where recognising that a face is present, and the owner of the face are treated as two separate problems. This line of questions also raises the issue of the quantity and quality of training material available.

5 EXAMPLE: ADVANCED COMPUTER GRAPHICS

A simpler but equally effective narrative is employed at the end of a final year computer graphics unit. A movie director tells the students that his new movie 'Rain of Terror: <insert city here> Drowns' has just been green-lit by the studio. He needs the students, as his visual effects team, to decide how to realise his vision of a number of key scenes. Most of the questions begin with the director's shot descriptions from the script. The questions are presented in the sequence they would appear following the story of the film.

6 LESSONS LEARNED

Scoring is not necessary to make a quiz format work and encourage the students to prepare answers. The sense of false jeopardy of the students preferred questions being answered provides sufficient motivation to do their best to prepare answers to all questions.

Active monitoring of the progress of the students during the preparation phase is necessary to ensure that the majority of groups have not become idle and begin to wander off topic. If most groups have answers to the majority of the questions it is safe to proceed. It's acceptable if the teams answering the last few questions do not have full answers as other groups tend to be happy to contribute if the session has remained active throughout.

7 CONCLUSIONS

The novelty of the use of a strong narrative helps generate immediate student engagement in the session. The agency given to the student groups in allowing them to select the order in which questions are answered further increases engagement that persists throughout the exercise. In the end of semester feedback the students frequently comment that they find these exercises both enjoyable and helpful to their understanding of the subject.

REFERENCES

- Jonathan Nolan. 2011. Person of Interest. First shown on CBS 22 September 2011.
 Learning on Screen. 2022. Box of Broadcast. <https://learningonscreen.ac.uk/ondemand>.
 Accessed: 2022-02-20.